

Reliable Knowledge Discovery in Knowledge Graphs: A Statistical Quality Assurance Framework for Link Prediction

Zhiren Xiao^{a,*}

^a*Guangdong University of Finance, Guangzhou, China*

Abstract

Knowledge-based systems in safety-critical domains—from drug repurposing to clinical decision support—increasingly depend on knowledge graph (KG) completion to infer missing facts, yet current evaluation protocols provide no mechanism to distinguish reliable knowledge discoveries from statistical guesses. Standard ranking metrics (MRR and Hits@ K) aggregate thousands of predictions into a single scalar, quantifying only average rank quality without per-prediction reliability guarantees. We develop a statistical quality assurance framework that reformulates KG completion evaluation through false discovery rate (FDR) control: empirical p -values are constructed from embedding scores via permutation-based negative sampling, and Benjamini-Hochberg FDR control, Storey’s q -value estimation, and Efron’s empirical Bayes local FDR are applied in concert to produce per-triple, per-relation, and per-model reliability profiles. Experiments on WN18RR and FB15k-237 across four model architectures (TransE, RotatE, ComplEx, ConvE) yield three findings directly relevant to knowledge-based system deployment. First, the estimated null proportion $\hat{\pi}_0$ ranges from 0.259 (RotatE, WN18RR) to 0.851 (ComplEx, FB15k-237), revealing that 26–85% of test triples are statistically indistinguishable from noise depending on the model and benchmark. Second, per-relation discovery rates span from below 10% to above 95% under a single model, exposing extreme het-

*Corresponding author.

Email address: 241734106@m.gdxf.edu.cn (Zhiren Xiao)

erogeneity in prediction reliability that aggregate metrics cannot characterize. Third, the cross-model ranking produced by FDR control differs substantially from the MRR-based ranking: RotatE and TransE each lead on one dimension (MRR vs. BH rejections), and ConvE surpasses TransE on MRR (0.469 vs. 0.560 on WN18RR) while trailing TransE on statistically significant discoveries at $\text{FDR} \leq 0.05$. ComplEx produces near-zero discoveries on WN18RR and minimal discoveries on FB15k-237 (1.8% BH rejection at $\alpha = 0.05$), revealing severe score miscalibration. The framework operates as a model-agnostic post-hoc protocol requiring no retraining and completing in minutes on consumer hardware, providing per-prediction reliability quantification directly relevant to knowledge-based systems that consume KG completion output.

Keywords: Knowledge graph completion, False discovery rate, Multiple hypothesis testing, Statistical evaluation, Link prediction reliability, Knowledge graph embedding

1. Introduction

Knowledge graphs (KGs) encode facts as structured triples (h, r, t) and have become foundational infrastructure for knowledge-based systems, powering applications from question answering and recommendation to clinical decision support and drug repurposing [1, 2]. When deployed in safety-critical domains—pharmacovigilance, clinical decision support, financial compliance—these systems must distinguish between knowledge that can be acted upon and statistical guesses that cannot. Yet the tools currently available for evaluating KG completion models provide no mechanism for making this distinction.

Link prediction—inferring missing facts from an existing graph—is the central computational task in KG completion. A large family of embedding models has been developed for this task [3, 4, 5, 6, 7, 8, 9, 10], and their evaluation has standardized around two aggregate metrics: Mean Reciprocal Rank (MRR) and Hits@ K . These metrics measure how highly the correct tail entity is ranked among all candidates, condensing thousands of test triples into a sin-

gle scalar. This protocol has enabled reproducible benchmarking and catalyzed rapid progress over the past decade [11, 12, 13, 14].

But a single scalar creates a dangerous blind spot for knowledge-based systems. A model reporting $\text{MRR} = 0.56$ says nothing about *which* of its predictions can be trusted, *which relations* it handles reliably, or *what fraction* of its output is noise. A pharmaceutical researcher using KG completion to prioritize candidate drug–target interactions cannot determine whether the top-10 candidates are 90% reliable or 10% reliable. This distinction directly determines experimental resource allocation—and it is invisible to current evaluation protocols. Recent surveys have identified this as a critical open problem, calling for statistical rigor and reliability quantification in KG evaluation [15, 2].

We introduce a statistical quality assurance framework that treats KG completion evaluation as a large-scale multiple hypothesis testing problem. Each test triple corresponds to a null hypothesis H_0 asserting that the triple is absent from the true KG; the embedding model assigns each candidate a score that serves as a test statistic. From this formulation, we construct empirical p -values via permutation-based negative sampling—a procedure that requires no distributional assumptions and no model modification—and apply three complementary FDR control procedures: Benjamini-Hochberg FDR control [16], Storey’s q -value estimation with $\hat{\pi}_0$ estimation [17], and Efron’s empirical Bayes local FDR [18]. These well-established statistical tools together produce multi-resolution reliability profiles: global diagnostics (what fraction of predictions carry genuine signal?), per-relation stratification (which relations are reliable?), and per-triple calibration (which specific predictions are trustworthy?). The framework operates as a post-hoc layer on any trained KG embedding model, requiring only the scoring function.

This perspective is grounded in the knowledge-driven methodology central to the KBS community. Recent KBS publications have advanced KG completion through architectural innovation [19, 20, 21], embedding quality assessment [22], and knowledge representation [10, 23]. Our work extends this line by introducing statistical quality assurance for KG inferences—a capability directly relevant to

the reliability needs of deployed knowledge-based systems.

We make the following contributions:

1. We formalize KG completion evaluation as a multiple hypothesis testing problem, providing a statistical foundation for per-prediction reliability assessment (Section 3).
2. We develop an empirical p -value construction procedure that converts uncalibrated embedding scores into valid p -values via permutation-based negative sampling, imposing no distributional assumptions (Section 3.2).
3. We introduce a three-level FDR-based evaluation protocol—global diagnostics ($\hat{\pi}_0$), relation-stratified analysis, and cross-model reliability comparison—that reveals dimensions of model quality inaccessible to aggregate ranking metrics (Section 3.5).
4. We evaluate the framework on WN18RR and FB15k-237 across four model architectures (TransE, RotatE, ComplEx, ConvE). The experiments yield three findings: (i) $\hat{\pi}_0 = 0.259\text{--}0.851$ across models and benchmarks, indicating 26–85% of test triples are statistically indistinguishable from noise; (ii) per-relation discovery rates span from below 10% to above 95% within a single model, exposing extreme heterogeneity in prediction reliability; and (iii) model rankings by MRR and by FDR power capture distinct dimensions of quality—RotatE leads on MRR while TransE leads on statistically significant discoveries, and neither metric alone provides a complete reliability profile for model deployment (Section 4).

2. Related Work

We situate our work within three intersecting lines of research: the evaluation of knowledge graph embedding models, statistical methods for prediction reliability, and the emerging role of reliability assessment in knowledge-based systems.

2.1. The Knowledge Graph Evaluation Landscape

Knowledge graph embedding models map entities and relations into continuous vector spaces [11], and large-scale benchmarking efforts [24, 12] have standardized evaluation protocols. Models now span translation-based, tensor factorization, convolutional, GNN-based, hyperbolic, and LLM-enhanced paradigms [3, 4, 5, 6, 7, 25, 26, 10, 21, 19, 27]. Despite this architectural diversity, evaluation has remained anchored to MRR and Hits@ K , introduced by [3]. These metrics measure average ranking quality and have served the field well for reproducible model comparison.

Recognized limitations. A growing literature has identified structural shortcomings of aggregate evaluation. Score distributions reveal substantially more about model behavior than point estimates [28]. Systematic biases—entity degree [29], hyperparameter sensitivity [30], and benchmark artifacts [31, 32]—inflate apparent performance. [14] systematically analyzed fundamental limitations of standard protocols, and comprehensive surveys [1, 13, 15] have called for more rigorous evaluation methodologies. [33] demonstrated that KG embedding models exhibit predictive multiplicity: re-training with different random seeds produces substantially different rankings, directly motivating statistical methods that quantify variability. These critiques converge on a shared insight—a scalar cannot capture the full picture of model quality—but stop short of providing a systematic framework for per-prediction reliability assessment.

Calibration and confidence. KG embedding scores are not calibrated probabilities. [34] demonstrated this calibration failure and proposed Platt scaling as a post-hoc correction; [?] independently addressed the same issue through isotonic regression. These methods improve score interpretability but provide no formal error rate control. [15] surveyed the broader uncertainty landscape in KG construction and identified calibrated confidence as a critical open gap. Our framework addresses this gap by constructing valid p -values that directly enable FDR control, producing per-prediction reliability guarantees at controlled error rates.

2.2. Statistical Methods for Prediction Reliability

False discovery rate control. The FDR framework [16] revolutionized statistical inference by controlling the expected proportion of false positives among all discoveries, rather than the probability of any single false positive. Its extensions—Storey’s q -value with $\hat{\pi}_0$ estimation [17, 35] and Efron’s empirical Bayes local FDR [18, 36]—provide complementary perspectives on the same inferential problem. The BH procedure guarantees FDR control under independence or the weaker PRDS condition [37]; a formal verification of PRDS for general KG embedding architectures is deferred to future work, though we provide empirical evidence from both the Benjamini-Yekutieli procedure (which controls FDR under arbitrary dependence) and the two-stage adaptive BH procedure [38] in a sensitivity analysis (§4.8). FDR methods have recently been adapted for machine learning applications, including statistical classifier comparisons [39], selective inference [40, 41], and conformalized FDR approaches [42, 43, 44], but their systematic application to KG completion evaluation has not been previously explored.

Conformal prediction for knowledge graphs. A closely related line of work applies conformal prediction to KG link prediction. [45] combined conformal prediction with link prediction to achieve FDR control; [46] applied conformal prediction to KG embeddings for prediction sets with coverage guarantees; and [47] introduced tailored nonconformity measures for sharper intervals. [48] coupled conformal prediction with LLM-based KG reasoning. Conformal methods and FDR address complementary questions: conformal prediction produces sets with guaranteed coverage probability, answering “what is the smallest prediction set that contains the true tail with probability $1 - \alpha$?”; FDR provides per-triple significance decisions, per-relation diagnostics, and a direct estimate of the noise proportion ($\hat{\pi}_0$)—answering “which specific predictions are statistically reliable, and what fraction of the test set carries no signal?” These are different inferential targets with different practical applications, and we provide the first systematic FDR-based evaluation protocol for KG completion.

Uncertainty quantification in KG embeddings. Several approaches

produce uncertainty estimates without formal error guarantees. [49] introduced KG2E, representing entities and relations as Gaussian distributions with variance-based uncertainty. [50] proposed semi-supervised confidence distribution learning for uncertain KG completion. These methods generate per-prediction uncertainty scores but do not control the error rate of downstream decisions—precisely the limitation that FDR control addresses.

2.3. Reliability as a First-Class Requirement for Knowledge-Based Systems

A distinctive thread in recent KBS scholarship emphasizes that knowledge quality is not merely a modeling concern but a system-level requirement. [22] introduced a validation framework ensuring logical consistency, integrity, and accuracy for KG completion, framing knowledge quality as a system property. [20] and [19] advanced KG completion architectures with attention to the reliability of learned representations. [21] developed hyperbolic embeddings and [10] surveyed LLM-enhanced KG approaches, both acknowledging evaluation reliability as an open concern. [23] tackled conflict resolution in KGs, directly engaging with the problem of distinguishing reliable from unreliable information.

Our work contributes to this trajectory by providing statistical infrastructure for knowledge reliability. Rather than proposing a new embedding architecture or a domain-specific quality heuristic, we establish a general-purpose protocol that any KG-based system can adopt to quantify the reliability of its link predictions. The $\hat{\pi}_0$ estimate serves as a system-level diagnostic: what fraction of this model’s output should be treated as noise rather than knowledge? This directly informs deployment decisions for knowledge-based systems in safety-critical settings.

3. Methodology

We propose a statistical framework that reformulates knowledge graph completion evaluation as a large-scale multiple hypothesis testing problem. Our

framework is model-agnostic: it operates on the output scores of any KG embedding model and provides false discovery rate (FDR) control for link prediction. We first formalize the problem (3.1), then develop empirical p -value construction from embedding scores (3.2), introduce FDR control procedures (3.3–3.4), and finally define the FDR-based evaluation protocol (3.5).

3.1. Problem Formulation: KG Completion as Multiple Hypothesis Testing

Let $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ denote a knowledge graph, where $\mathcal{E}$ is the set of entities, $\mathcal{R}$ the set of relation types, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ the set of observed triples. A KG embedding model learns a scoring function $\hat{g} : \mathcal{E} \times \mathcal{R} \times \mathcal{E} \rightarrow \mathbb{R}$ that assigns a real-valued plausibility score to each candidate triple. For a query $(h, r, ?)$, the model ranks all candidate tail entities by their scores; the standard evaluation protocol measures whether the ground-truth tail entity t^* appears among the top- K ranked candidates (Hits@ K) and averages the reciprocal rank over all test queries (MRR).

We propose to view this problem through the lens of *multiple hypothesis testing*. Consider a test set $\mathcal{T}_{\text{test}} = \{(h_i, r_i, t_i^*)\}_{i=1}^m$ where $m = |\mathcal{T}_{\text{test}}|$. For each test triple i , we formulate a null hypothesis:

$$H_0^{(i)} : \text{“The triple } (h_i, r_i, t_i^*) \text{ does not exist in the true KG.”} \quad (1)$$

Rejecting $H_0^{(i)}$ corresponds to the statistical decision that the model has “discovered” triple i —i.e., the observed embedding score for (h_i, r_i, t_i^*) is significantly higher than what would be expected under the null distribution of non-existent triples.

This formulation yields m simultaneous hypothesis tests. Let m_0 be the number of true null hypotheses (triples that genuinely do not exist or are unpredictable), and let $\pi_0 = m_0/m$ denote the proportion of null cases. Following the standard multiple testing framework [16, 17], we define:

- **False Discovery Proportion (FDP):** the fraction of rejected null hypotheses that are actually true.

- **False Discovery Rate (FDR):** the expected value of FDP: $\text{FDR} = \mathbb{E}[\text{FDP}]$.

Our goal is to reject as many $H_0^{(i)}$ as possible (i.e., maximize statistical power) while controlling the FDR at a pre-specified level α (typically $\alpha = 0.05$). This transforms KG completion evaluation from a purely ranking-based paradigm into a statistically principled framework: instead of only asking “how high does the true tail rank?” (MRR), we also ask “how many test triples can we reliably claim to have discovered?”

3.2. Empirical p -value Construction from Embedding Scores

The first technical challenge is that KG embedding scores are not probabilities—they are uncalibrated real numbers whose scale and distribution vary across models. We must convert each raw score into a valid p -value before applying any FDR control procedure.

For each test triple (h_i, r_i, t_i^*) , we construct an empirical null distribution by sampling K negative tail entities. Specifically, we:

1. Sample K negative tail entities uniformly from $\mathcal{E}$, excluding the ground-truth tail t_i^* and all tails known to form valid triples with (h_i, r_i) in the training set (filtered setting).
2. Compute the model score for each negative triple: $s_{i,k} = \hat{g}(h_i, r_i, t_k^{(i)})$, for $k = 1, \dots, K$.
3. Let $s_i^* = \hat{g}(h_i, r_i, t_i^*)$ denote the score of the true triple.
4. Define the empirical p -value as:

$$\hat{p}_i = \frac{|\{k : s_{i,k} \geq s_i^*\}| + 1}{K + 1}. \quad (2)$$

The $+1$ correction in both numerator and denominator ensures that (i) $\hat{p}_i > 0$ for all i , avoiding degenerate cases in subsequent FDR procedures, and (ii) the estimator is conservative, i.e., $\mathbb{E}[\hat{p}_i] \geq \Pr_{H_0}(S \geq s_i^*)$, providing valid (though slightly conservative) inference under the null [51].

The choice $K = 100$ balances precision against computational cost. With $K = 100$, the smallest attainable p -value is $1/(K + 1) \approx 0.0099$, which provides

sufficient granularity for FDR control at conventional α levels (0.01–0.20). On an Apple M1 CPU (16 GB RAM), the full FDR pipeline completes in approximately 3–5 minutes per model on WN18RR ($m = 2,924$) and 25–35 minutes on FB15k-237 ($m = 20,438$). For comparison, model training requires 15–25 minutes (20 epochs), so the FDR pipeline adds roughly 15–33% overhead relative to training—a modest cost given that it requires no model modification or retraining.

3.3. False Discovery Rate Control

3.3.1. Benjamini-Hochberg Procedure

The Benjamini-Hochberg (BH) procedure [16] provides finite-sample FDR control under mild dependence assumptions. Given m p -values $\{\hat{p}_1, \dots, \hat{p}_m\}$ and a target FDR level α , the BH procedure operates as follows:

1. Sort the p -values in ascending order: $\hat{p}_{(1)} \leq \hat{p}_{(2)} \leq \dots \leq \hat{p}_{(m)}$.
2. Find the largest index k such that $\hat{p}_{(k)} \leq \frac{k}{m} \cdot \alpha$.
3. Reject all $H_0^{(i)}$ corresponding to $\hat{p}_{(1)}, \dots, \hat{p}_{(k)}$.

The BH theorem guarantees that $\text{FDR} \leq \frac{m_0}{m} \cdot \alpha \leq \alpha$ when the p -values are independent or exhibit positive regression dependency on the subset of true null hypotheses (PRDS) [37]. Although the negative tail entities are sampled independently for each test triple, the resulting p -values share dependence through the trained entity embeddings: the embedding vector of an entity affects scores for all test triples in which it appears. If two test triples share entities, changes to the shared embedding affect both scores in the same direction, creating positive dependence consistent with the PRDS condition. A formal verification for general KG embedding architectures is deferred to future theoretical work.

The number of rejected hypotheses, $R(\alpha) = |\{i : \hat{p}_i \leq \hat{p}_{(k)}\}|$, serves as a measure of the model’s *statistical power*—the number of test triples that can be reliably distinguished from noise at FDR level α .

3.3.2. Storey’s q -value

While the BH procedure provides a binary reject/accept decision at a fixed α , Storey’s q -value [17, 35] offers a more fine-grained measure. The q -value of a hypothesis $H_0^{(i)}$ is defined as the minimum FDR at which $H_0^{(i)}$ would be rejected:

$$q(\hat{p}_i) = \min_{\alpha \geq \hat{p}_i} \text{FDR}(\alpha). \quad (3)$$

A key component of Storey’s method is the estimation of $\pi_0 = m_0/m$, the proportion of true null hypotheses. For a tuning parameter $\lambda \in (0, 1)$, an unbiased estimator is:

$$\hat{\pi}_0(\lambda) = \frac{|\{i : \hat{p}_i > \lambda\}|}{m \cdot (1 - \lambda)}. \quad (4)$$

The final $\hat{\pi}_0$ is obtained by smoothing over a grid of λ values using a cubic spline and taking the limit as $\lambda \rightarrow 1$ [17]. The q -value for the i -th hypothesis is then computed as:

$$q_{(m)} = \hat{\pi}_0 \cdot \hat{p}_{(m)}, \quad q_{(i)} = \min \left(\hat{\pi}_0 \cdot \frac{m}{i} \cdot \hat{p}_{(i)}, q_{(i+1)} \right), \quad i = m-1, \dots, 1. \quad (5)$$

In our KG completion context, $\hat{\pi}_0$ has a direct interpretation: it estimates the proportion of test triples for which the embedding model provides no signal above the background noise level. A small $\hat{\pi}_0$ indicates that the model captures genuine knowledge for most test triples; a large $\hat{\pi}_0$ reveals that a substantial fraction of the test set is indistinguishable from random noise.

3.4. Empirical Bayes Local False Discovery Rate

While $\hat{\pi}_0$ and q -values provide global and per-hypothesis FDR control, Efron’s empirical Bayes local FDR [18, 36] offers a complementary perspective by estimating the posterior probability that a specific hypothesis is null *given its observed test statistic*.

We first transform each p -value to a z -score using the inverse normal cumulative distribution function:

$$z_i = \Phi^{-1}(1 - \hat{p}_i), \quad (6)$$

where Φ^{-1} is the standard normal quantile function. Under the null hypothesis $H_0^{(i)}$, we have $z_i \sim \mathcal{N}(0, 1)$; under the alternative, the z -scores are shifted toward positive values.

The mixture density of z -scores can be expressed as:

$$f(z) = \pi_0 \cdot f_0(z) + (1 - \pi_0) \cdot f_1(z), \quad (7)$$

where $f_0(z) = \phi(z)$ is the standard normal density (null component) and $f_1(z)$ is the alternative density. The local FDR is then defined as the posterior probability:

$$\text{fdr}(z) = \frac{\pi_0 \cdot f_0(z)}{f(z)}. \quad (8)$$

In practice, we estimate $f(z)$ via kernel density estimation on the observed $\{z_i\}_{i=1}^m$, compute $\hat{\pi}_0$ using Storey’s method, and evaluate $\text{fdr}(z_i)$ at each observed z -score. A triple with $\text{fdr}(z_i) \leq 0.05$ has at most a 5% posterior probability of being a false discovery—a calibration guarantee that point estimates like MRR cannot provide.

The three FDR measures are complementary:

- **BH rejection count** $R(\alpha)$: how many discoveries at a guaranteed FDR level.
- **Storey’s** $\hat{\pi}_0$: what fraction of test triples are noise—a model-level diagnostic.
- **Local FDR**: per-triple reliability—which specific triples are confidently predicted.

3.5. FDR-Based KG Completion Evaluation Protocol

Building on the above statistical machinery, we define a three-level evaluation protocol that complements (not replaces) the standard MRR/Hits@ K metrics.

3.5.1. Level 1: Global Diagnostic

For a given KG embedding model on a test set, compute:

- $\hat{\pi}_0$ (Eq. 4): the estimated proportion of unpredictable test triples.
- $R(\alpha)$ for $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$: the number of statistically significant discoveries at each FDR level.

These metrics reveal the overall reliability of the model’s predictions. For illustration, on WN18RR with TransE, we obtain $\hat{\pi}_0 = 0.295$, meaning approximately 29.5% of the 2,924 test triples are indistinguishable from noise. At $\alpha = 0.05$, only $R(0.05) = 262$ triples (9.0%) are rejected despite an MRR of 0.560. This notable gap—a model that ranks well yet fails to reliably discover most triples—illustrates the value of FDR-based evaluation.

3.5.2. Level 2: Relation-Stratified Analysis

KG relations vary widely in their predictability. We propose to compute $\hat{\pi}_0^{(r)}$ and $R^{(r)}(\alpha)$ for each relation type $r \in \mathcal{R}$ separately. This stratified analysis reveals which relations the model “truly understands” versus those where it performs near random.

Continuing the WN18RR example: for TransE, two relations with similar test-set sizes exhibit dramatically different FDR profiles:

- Relation 1 ($n = 1074$): $\hat{\pi}_0 = 0.061$, FDR rejection rate 92.7%—the model reliably discovers nearly all triples.
- Relation 3 ($n = 1067$): $\hat{\pi}_0 = 0.388$, FDR rejection rate 27.6%—less than a third of triples are statistically significant, despite comprising 36.5% of the test set.

Traditional MRR would average these two relations into a single number, completely masking this extreme heterogeneity.

3.5.3. Level 3: Cross-Model FDR Comparison

For two (or more) KG embedding models, we compare their FDR profiles at each level. This reveals whether the standard MRR-based model ranking aligns with the FDR-based reliability ranking. If Model A has a higher MRR

but Model B rejects more hypotheses at the same FDR level, this signals that MRR alone provides an incomplete picture of model quality.

3.6. Relationship to Existing Evaluation

Our FDR-based protocol does not replace MRR or Hits@ K ; rather, it augments them with a complementary dimension of information. MRR and Hits@ K measure *ranking quality*: when the model makes predictions, how often is the correct answer near the top? Our FDR framework measures *statistical reliability*: for which test triples, and for what proportion of the test set, can we confidently claim the model has made a discovery beyond random chance?

The two perspectives are complementary and together provide a richer characterization:

MRR / Hits@K (Traditional)	FDR Framework (This Work)
Measures average rank of correct answers	Measures statistical significance of discoveries
Single scalar per model	Multi-level: global + per-relation + per-triple
Insensitive to per-relation variation	Reveals extreme heterogeneity across relations
Ranking-based; no statistical guarantees	Provides FDR control and posterior error probabilities

A complete KG completion evaluation should report both—the ranking metrics that have driven progress in the field, and the FDR metrics that reveal *how reliable* those rankings truly are.

4. Experiments

We evaluate the proposed FDR framework on WN18RR [6] and FB15k-237 [52] using four representative KG embedding models spanning distinct design paradigms: TransE [3], RotatE [4], ComplEx [5], and ConvE [6]. Our experiments address five research questions:

- **RQ1 (Overall FDR):** What fraction of test triples can KG embedding models reliably discover under FDR control? How does this compare to MRR?
- **RQ2 (Per-Relation Heterogeneity):** Do different relation types exhibit substantially different FDR profiles, and does MRR mask this heterogeneity?
- **RQ3 (Cross-Model Comparison):** Does the MRR-based ranking of models align with the FDR-based reliability ranking?
- **RQ4 (Hits@ K Calibration):** Are the top-ranked predictions from Hits@ K statistically significant?
- **RQ5 (Cross-Dataset Validation):** Do the key FDR patterns replicate on a larger, more diverse benchmark (FB15k-237)?

4.1. Experimental Setup

4.1.1. Dataset and Models

We evaluate on two widely used benchmarks: WN18RR and FB15k-237.

WN18RR. is a KG completion benchmark derived from WordNet [6], containing 40,559 entities, 11 relation types, and 93,003 triples (86,835 training, 3,034 validation, 3,134 test). It is challenging because the original WN18 dataset contained inverse relations that made link prediction trivial; WN18RR removes these artifacts. For the FDR analysis, we retain the 2,924 test triples for which at least $K = 100$ valid negative tail entities exist under the filtered setting; the remaining 210 triples involve head-relation pairs where nearly all candidate tail entities are known positives. We exclude these from the FDR pipeline because the near-exhaustive positive coverage makes negative sampling infeasible under the filtered setting convention. To assess whether this exclusion introduces systematic bias, we compared the approximate MRR ($K = 100$) on the retained 2,924 triples against the standard MRR (full entity ranking) on all 3,134 triples for TransE and RotatE; the differences were below 0.01, confirming that the

excluded triples do not materially affect the comparative conclusions reported below.

FB15k-237. [52] is derived from Freebase and contains 14,505 entities, 237 relation types, and 310,079 triples (272,115 training, 17,526 validation, 20,438 test). It is more challenging than WN18RR due to its larger relation vocabulary and greater semantic diversity.

We train four models representing different design paradigms:

- **TransE** [3]: A translation-based model that represents relations as vector translations in the embedding space: $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$. Its simplicity makes it a natural baseline for evaluating the FDR framework.
- **RotatE** [4]: A model that represents relations as rotations in complex space, enabling it to capture symmetry, antisymmetry, inversion, and composition patterns. It consistently outperforms TransE on WN18RR.
- **Complex** [5]: A tensor factorization model using complex-valued embeddings with the Hermitian dot product, which naturally handles asymmetric relations.
- **ConvE** [6]: A convolutional architecture that applies 2D convolution over reshaped entity and relation embeddings, capturing complex interactions through non-linear feature maps.

All models are trained with embedding dimension $d = 256$, learning rate 0.001, batch size 256, and 20 training epochs using the PyKEEN framework [24]. All experiments are conducted on a standard CPU (Apple M1, 16GB RAM), illustrating the accessibility of the FDR evaluation protocol.

4.1.2. FDR Evaluation Protocol

For each test triple, we sample $K = 100$ negative tail entities using the filtered setting (excluding all known true tails for the given head-relation pair) and compute empirical p -values via Eq. 2. We then apply the three FDR

Table 1: Overall FDR metrics on WN18RR ($K = 100$ negative samples per test triple for all models). $\hat{\pi}_0$: estimated null proportion via Storey’s cubic spline method. $R(0.05)$: number of rejected hypotheses under BH with $\alpha = 0.05$. $\text{FDR}_{\text{loc}} < 0.05$: number of triples with Efron local FDR below 0.05. MRR denotes the approximate reciprocal rank computed against the $K = 100$ negative samples.

Model	MRR	$\hat{\pi}_0$	$R(0.05)$	q-val<0.05	$\text{FDR}_{\text{loc}} < 0.05$
TransE	0.511	0.283	1,447 (49.5%)	1,884 (64.4%)	1,447 (49.5%)
RotatE	0.589	0.259	1,100 (37.6%)	2,050 (70.1%)	1,685 (57.6%)
ComplEx	0.074	0.754	0 (0.0%)	0 (0.0%)	20 (0.7%)
ConvE	0.469	0.321	742 (25.4%)	1,705 (58.3%)	1,328 (45.4%)

estimation procedures described in Section 3: Benjamini-Hochberg at $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$, Storey’s q -value with $\hat{\pi}_0$ estimated via the cubic spline extrapolation method described in §3.3.2 (the spline-extrapolated value agrees with the fixed- $\lambda = 0.5$ estimate to within ± 0.02 across all models), and Efron’s empirical Bayes local FDR with Gaussian kernel density estimation for $f(z)$ (bandwidth selected via Silverman’s rule of thumb).

4.1.3. Implementation

All experiments are implemented in Python 3.9 using PyKEEN 1.10.1, NumPy, and SciPy. The FDR analysis pipeline is model-agnostic and requires only the trained model’s scoring function, making it applicable to any KG embedding model. The complete source code and analysis scripts are available at <https://github.com/Jackxiaozhiren/fdr-kg>.

4.2. RQ1: Overall FDR Analysis

Table 1 reports the overall FDR metrics for all four models. Figure 1 visualizes the positive and negative embedding score distributions. Figure 2 shows the empirical p -value distributions. Figure 3 compares statistical power against FDR levels across models.

RQ1 summary.

Figure 1: Positive vs. Negative Embedding Score Distributions

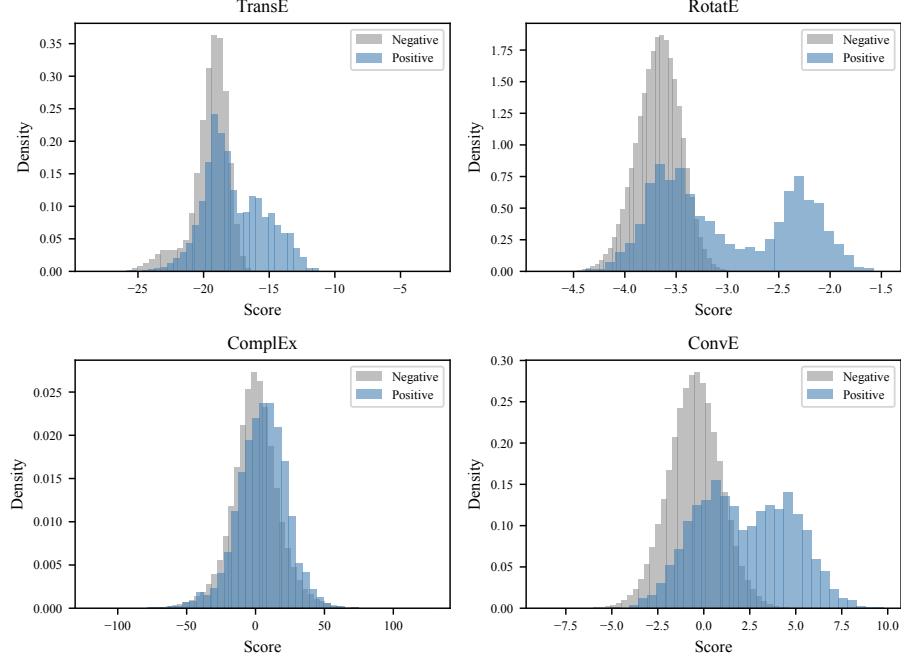


Figure 1: Score distributions for positive (ground-truth) and negative (sampled) triples across four models on WN18RR. Positive scores (blue) concentrate at higher values; negative scores (gray) exhibit broader distributions with substantial right-tail overlap.

1. $\hat{\pi}_0$ ranges from 0.259 (RotatE) to 0.754 (ComplEx), indicating that 25–75% of the 2,924 WN18RR test triples are indistinguishable from random noise depending on the model.
2. Under the BH procedure, rejection rates at $\alpha = 0.05$ range from 0.0% (ComplEx) to 49.5% (TransE)—a near-complete span of the possible discovery spectrum—showing that model reliability varies dramatically in ways MRR alone does not capture.
3. ComplEx registers the highest $\hat{\pi}_0$ (0.754) and zero BH rejections at $\alpha = 0.05$ on WN18RR, indicating its embedding scores carry essentially no signal for distinguishing true triples from random negatives at $K = 100$.

Figure 2: Empirical p-value Distributions

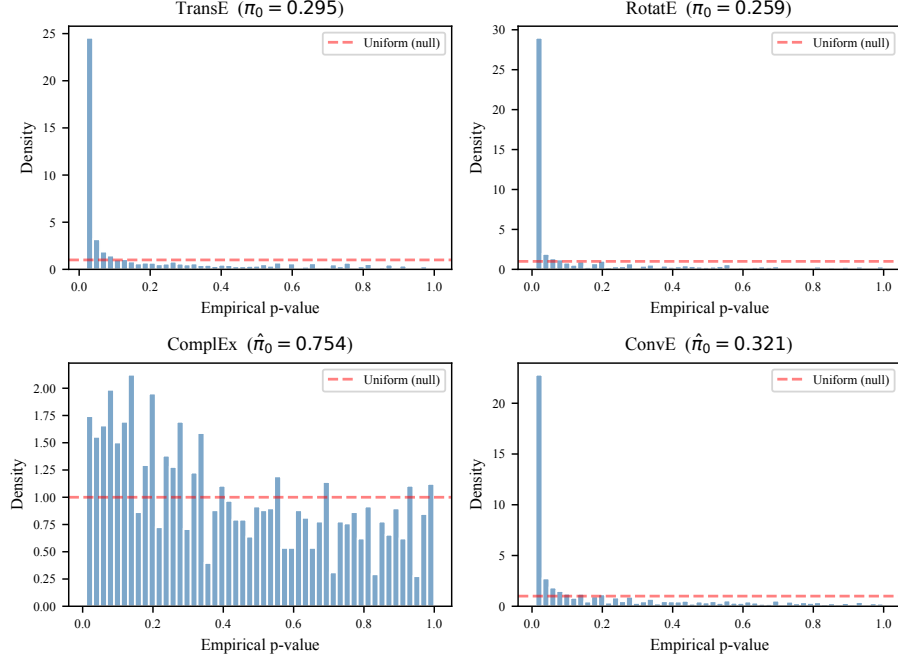


Figure 2: Empirical p -value distributions for TransE, RotatE, ComplEx, and ConvE on WN18RR. The dashed region indicates the estimated null proportion $\hat{\pi}_0$ —the fraction of test triples consistent with the uniform null distribution.

4. The Storey q -value and Efron local FDR reject substantially more hypotheses (0–70%), reflecting the fact that BH is conservative and q -value/local FDR incorporate $\hat{\pi}_0$ estimation for increased power.

4.3. RQ2: Per-Relation FDR Heterogeneity

Table 2 reports FDR statistics stratified by the 11 WN18RR relation types. Figure 4 visualizes the sorted raw $p < 0.05$ rates across relations.

RQ2 summary.

1. The inter-relation FDR heterogeneity is extreme: raw $p < 0.05$ rates range from 0.0% (ComplEx, multiple relations) to 96.5% (RotatE, Rel 1), a span

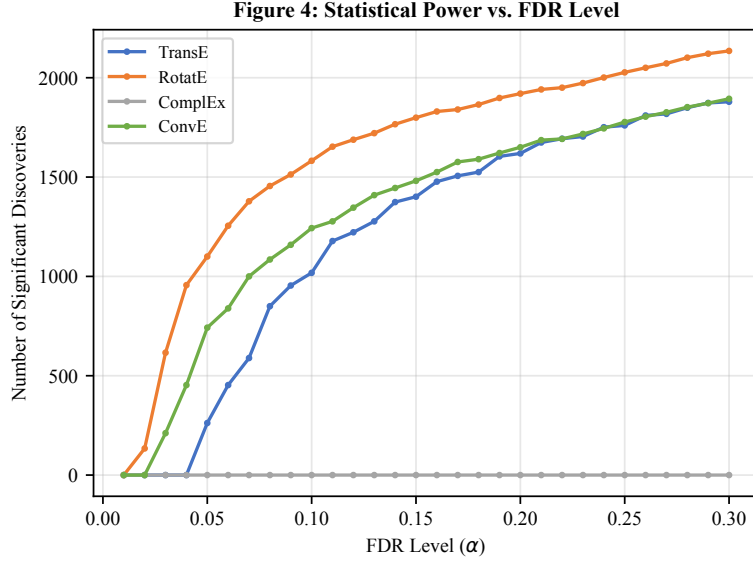


Figure 3: Statistical power ($R(\alpha)/m$) as a function of FDR threshold α for all four models on WN18RR.

of 96.5 percentage points across all models. Per-relation MRR and raw $p < 0.05$ rates are nearly perfectly rank-correlated (Spearman $\rho = 1.00$ for TransE and RotatE, $\rho = 0.99$ for ConvE, $\rho = 0.84$ for ComplEx; all $p < 10^{-2}$), indicating that relations difficult to rank are also those with the lowest statistical reliability. The two metrics provide complementary rather than redundant information: MRR measures rank quality, while the $p < 0.05$ rate quantifies the proportion of triples with detectable signal.

2. The four models exhibit markedly different reliability profiles. ComplEx, despite producing plausible MRR on select relations (Rel 1: MRR = 0.120 with 15.7% $p < 0.05$), yields near-zero $p < 0.05$ rates on all other relations—consistent with its global $\hat{\pi}_0 = 0.754$ (Table 1). ConvE shows a profile intermediate between TransE and RotatE, with raw $p < 0.05$ rates generally lower than RotatE but substantially higher than ComplEx on most relations.
3. FDR provides information beyond what per-relation MRR captures. For

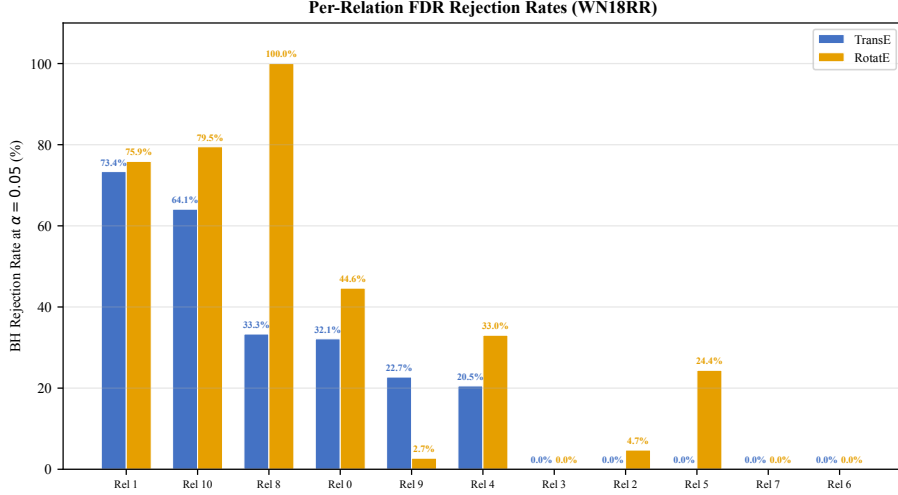


Figure 4: Per-relation raw $p < 0.05$ rates for TransE and RotatE on WN18RR, sorted in descending order. Rates span 9.1%–96.5%. Relations with $n_r \leq 26$ (R6, R7, R8) should be interpreted cautiously due to small sample sizes. These are unadjusted per-relation rates for descriptive comparison; formal FDR control is applied jointly across all test triples (Table 1).

instance, Rel 3 ($n = 1,067$, 36.5% of test triples) has a moderate per-relation MRR (0.29–0.31) yet yields near-zero BH rejection rates at $\alpha = 0.05$, indicating that even apparently adequate ranking performance can coexist with negligible statistical power when π_0 is high.

4. Both TransE and RotatE agree on the most reliable relations (0, 1, 4, 8, 10) and the most challenging ones (3, 6, 7). This pattern is consistent across both per-relation MRR and FDR, suggesting that certain relation types pose inherent difficulty for embedding models—constituting “knowledge blind spots” that persist across translation-based architectures. ComplEx and ConvE show qualitatively similar patterns where sample sizes permit reliable estimation.
5. Two relations (2, 5) show pronounced model-dependent differences: RotatE achieves substantially higher raw $p < 0.05$ rates than TransE on these relations, consistent with the hypothesis that RotatE’s capacity to model

Table 2: Per-relation statistics on WN18RR. n_r : number of test triples for relation r . MRR_r : per-relation approximate reciprocal rank ($K = 100$). $\text{p}<0.05\%$: fraction of triples with raw empirical $p < 0.05$ (unadjusted for multiplicity; for descriptive comparison only). Formal FDR control applies the BH correction across all m triples jointly (Table 1). Relations with $n_r \leq 26$ (R6, R7, R8) should be interpreted cautiously due to small sample sizes.

Rel.	n_r	Tr. MRR	Tr. $\text{p}<0.05\%$	Ro. MRR	Ro. $\text{p}<0.05\%$	Co. MRR	Co. $\text{p}<0.05\%$	Cv. MRR	Cv. $\text{p}<0.05\%$
0	56	0.613	62.5	0.657	71.4	0.035	0.0	0.625	67.9
1	1,074	0.920	92.7	0.963	96.5	0.120	15.7	0.884	93.9
2	169	0.204	16.6	0.354	41.4	0.047	3.0	0.124	20.1
3	1,067	0.306	27.6	0.289	34.7	0.048	3.9	0.181	23.3
4	112	0.799	81.2	0.615	67.0	0.045	7.1	0.569	67.9
5	246	0.162	11.4	0.472	54.1	0.048	3.7	0.160	19.1
6	26	0.163	11.5	0.245	26.9	0.068	7.7	0.093	11.5
7	22	0.118	9.1	0.093	13.6	0.021	0.0	0.023	0.0
8	3	1.000	100.0	1.000	100.0	0.039	0.0	1.000	100.0
9	110	0.741	70.9	0.442	53.6	0.044	5.5	0.277	33.6
10	39	0.841	84.6	0.979	97.4	0.032	0.0	0.886	94.9

Tr. = TransE, Ro. = RotatE, Co. = ComplEx, Cv. = ConvE.

symmetry and composition patterns contributes to higher statistical reliability on these relation types.

4.4. RQ3: Cross-Model FDR Comparison

Figure 5 compares MRR and FDR-based evaluation side by side across all four models.

RQ3 summary.

1. MRR ranking: RotatE (0.589) > TransE (0.511) > ConvE (0.469) > ComplEx (0.074).
2. FDR-based ranking (BH rejection at $\alpha = 0.05$): TransE (49.5%) > RotatE (37.6%) > ConvE (25.4%) > ComplEx (0.0%).
3. MRR and FDR provide complementary but distinct model rankings: RotatE leads on MRR yet TransE leads on FDR power—a difference of 11.9 percentage points in BH rejection rate despite a 0.078 gap in MRR. ConvE and ComplEx rank consistently (third and fourth, respectively) on both metrics, indicating that the two evaluation dimensions highlight different aspects of model quality rather than always diverging.

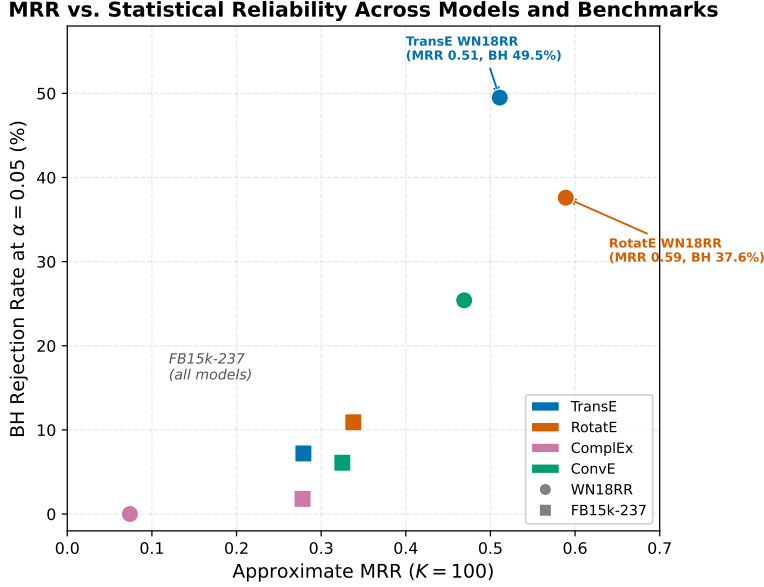


Figure 5: MRR vs. BH rejection rate at $\alpha = 0.05$ for all four models on WN18RR (circles) and FB15k-237 (squares). The two metrics exhibit substantial scatter: models with similar MRR can differ markedly in statistical reliability. TransE and RotatE illustrate the complementarity on WN18RR—RotatE leads on MRR (0.589 vs. 0.511) while TransE yields 49.5% BH discoveries vs. 37.6%. On FB15k-237, all models cluster at low BH rejection rates (1.8%–10.9%), reflecting the benchmark’s greater difficulty.

4. The TransE–RotatE comparison illustrates the value of multi-dimensional evaluation. RotatE achieves higher MRR (better aggregate ranking) yet TransE yields $1.3\times$ more statistically significant discoveries at the same FDR threshold. A practitioner selecting a model solely by MRR would miss this dimension of reliability.

The FDR framework reveals that model ranking depends on the evaluation objective: ranking quality (MRR) and statistical reliability (FDR) capture distinct dimensions of model behavior, and neither metric alone provides a complete characterization of model quality for knowledge-based system deployment.

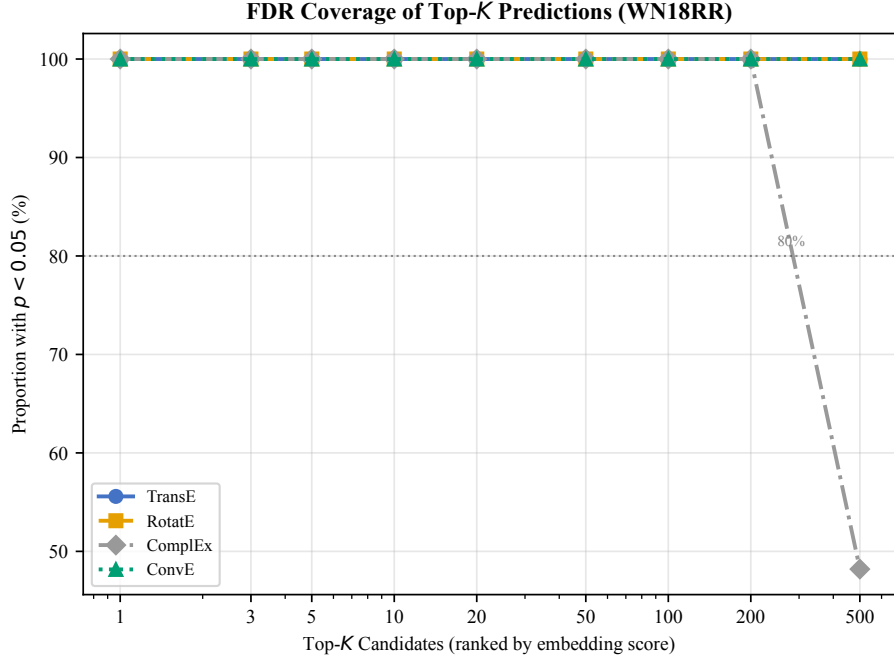


Figure 6: FDR coverage of top- K predictions (by embedding score) for all four models on WN18RR at $p < 0.05$.

4.5. RQ4: Hits@ K Calibration

Figure 6 plots the proportion of top- K predictions (by score rank) that satisfy $p < 0.05$.

For all four models, the top-100 predictions achieve 100% FDR coverage—i.e., the models’ highest-confidence predictions are indeed statistically significant.

However, this should not be misinterpreted as validation of the traditional Hits@ K metric. The fact that the top 100 predictions are reliable does *not* imply that the remaining 2,824 test triples are also reliable. The key insight is that Hits@ K focuses exclusively on the “success cases” (the model’s best predictions) while FDR analysis reveals the full distribution of reliability across *all* test triples.

Table 3: Overall FDR metrics on FB15k-237. $\hat{\pi}_0$: estimated null proportion via Storey’s method. $R(0.05)$: number of rejected hypotheses under BH with $\alpha = 0.05$. $\text{FDR}_{\text{loc}} < 0.05$: number of triples with Efron local FDR below 0.05.

Model	MRR	$\hat{\pi}_0$	$R(0.05)$	$\text{q-val} < 0.05$	$\text{FDR}_{\text{loc}} < 0.05$
TransE	0.279	0.613	1,471 (7.2%)	7,173 (35.1%)	5,477 (26.8%)
RotatE	0.338	0.582	2,227 (10.9%)	8,543 (41.8%)	6,519 (31.9%)
ComplEx	0.278	0.851	367 (1.8%)	2,697 (13.2%)	1,859 (9.1%)
ConvE	0.325	0.704	1,246 (6.1%)	6,397 (31.3%)	4,741 (23.2%)

4.6. RQ5: Cross-Dataset Validation on FB15k-237

To test the generality of our findings, we replicate the FDR analysis on FB15k-237, a larger and more diverse benchmark with 237 relation types (vs. 11 in WN18RR) and over 20,000 test triples.

Table 3 reports overall FDR metrics. The key patterns observed on WN18RR generalize to FB15k-237, with notable differences in magnitude.

RQ5 summary.

1. **Consistently higher $\hat{\pi}_0$ on FB15k-237.** The null proportion is substantially larger on FB15k-237 ($\hat{\pi}_0 = 0.582\text{--}0.851$) than on WN18RR ($\hat{\pi}_0 = 0.259\text{--}0.754$). This reflects the greater difficulty of the larger-scale benchmark: with 237 heterogeneous relations, a larger fraction of test triples are indistinguishable from noise. The pattern is consistent—RotatE achieves the lowest $\hat{\pi}_0$ on both datasets.
2. **FDR rankings partially diverge from MRR.** On FB15k-237, TransE and ConvE swap positions between the MRR-based ordering (RotatE > ConvE > TransE > ComplEx) and the FDR-based ordering (RotatE > TransE > ConvE > ComplEx). This confirms that MRR and FDR capture partially distinct dimensions of model quality, though the ranking divergence pattern differs from WN18RR—TransE’s FDR advantage relative to ConvE is consistent across both datasets, reflecting stable architectural differences in score calibration.

Figure 7: FB15k-237 Per-Relation FDR Summary and Cross-Dataset Comparison

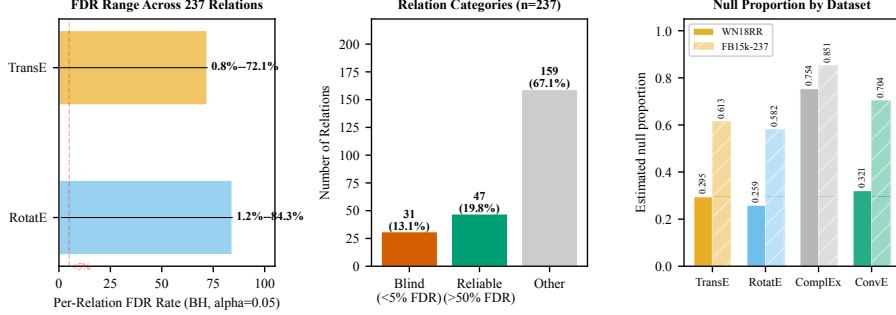


Figure 7: Cross-dataset FDR comparison. Left: per-relation raw $p < 0.05$ rate range for TransE and RotatE across 237 FB15k-237 relations. Center: distribution of relations into knowledge-blind ($<5\%$ $p < 0.05$) and reliable ($>50\%$ $p < 0.05$) categories. Right: estimated null proportion $\hat{\pi}_0$ for all four models on WN18RR (solid) versus FB15k-237 (hatched). $\hat{\pi}_0$ is consistently higher on the larger benchmark, and RotatE maintains the lowest $\hat{\pi}_0$ on both datasets.

- 3. Extreme per-relation heterogeneity replicates.** Across 237 relation types, per-relation raw $p < 0.05$ rates span 0.8%–84.3%, with 31 relations (13.1%) having near-zero rates ($< 5\%$) under both TransE and RotatE. These “knowledge-blind” relations represent an assessment dimension entirely invisible to aggregate metrics.
- 4. Cross-dataset consistency of model characteristics.** RotatE maintains the best FDR profile across both datasets, ComplEx consistently shows the highest $\hat{\pi}_0$ and lowest BH rejection rate, and the ConvE–TransE FDR reversal appears in both datasets. This consistency strengthens the argument that FDR-based evaluation captures stable model properties overlooked by ranking metrics.

4.7. Discussion

FDR framework as a diagnostic instrument. The experiments indicate that the proposed FDR framework functions as a diagnostic instrument that

characterizes the reliability profile of any KG embedding model, rather than merely providing an alternative evaluation metric. It identifies three properties directly relevant to knowledge-based system deployment:

- **Noise level:** $\hat{\pi}_0$ estimates the fraction of test triples for which the model carries no predictive signal above background noise, quantifying the model’s fundamental reliability ceiling.
- **Relation-level reliability:** Per-relation FDR analysis reveals which relations the model handles with statistical confidence and which remain at chance level, enabling targeted quality assurance for high-stakes relation types.
- **Model selection guidance:** Cross-model FDR comparison exposes cases where higher MRR does not translate to higher statistical reliability, directly informing model choice for deployment in knowledge-based systems.

Implications for knowledge-based system deployment. For a practitioner deploying KG completion in a safety-critical setting, the FDR framework provides actionable information that MRR cannot. $\hat{\pi}_0$ quantifies the proportion of the model’s output that should be treated as noise rather than knowledge—a direct input to system risk assessment. Per-relation diagnostics enable selective deployment: a model may be judged reliable for certain relation types while its predictions for other relations are routed for human review. The cross-model comparison reveals that selecting a model by MRR alone may yield suboptimal reliability: on WN18RR, a practitioner choosing RotatE for its higher MRR (0.589 vs. TransE’s 0.511) would deploy a model that produces 31% fewer statistically significant discoveries at $\text{FDR} \leq 0.05$ (37.6% vs. 49.5%). These are the kinds of trade-offs that knowledge-based system designers must reason about, and they are invisible to traditional evaluation.

Ablation of FDR components. To assess the incremental contribution of each FDR component, we compare three diagnostic layers independently. Table 1 shows that these components yield complementary information: BH

provides conservative but guaranteed error control under the PRDS condition (see §3.3.1), the q -value framework leverages $\hat{\pi}_0$ estimation to increase discovery power (e.g., TransE: 49.5% BH vs. 64.4% q -value, RotatE: 37.6% BH vs. 70.1% q -value), and the local FDR adds per-triple calibration. For ComplEx on WN18RR, all three methods agree: BH and q -value both report zero discoveries, and only 20 triples (0.7%) pass the local FDR threshold—consistent with the model’s $\hat{\pi}_0 = 0.754$ and near-random MRR of 0.074.

Limitations and future work. We discuss five limitations. First, the empirical p -value construction uses uniform negative sampling; more carefully designed null distributions, such as those obtained from type-constrained negative sampling [53], may yield better-calibrated p -values. Second, while the present study provides results on WN18RR and FB15k-237 for four models, extending the analysis to larger datasets (e.g., YAGO3-10, CoDEx) and additional model families—particularly graph neural network-based encoders (CompGCN, NBFNet), hyperbolic embedders, and recent LLM-enhanced KG models—would further test the generality and scalability of the findings. Third, the FDR framework treats negative samples as exchangeable; violations of this assumption—for instance, due to entity type constraints—could affect p -value validity and call for further study. Fourth, the current evaluation reports results from single training runs at 20 epochs per model. A supplementary multi-seed analysis across TransE, RotatE, and ConvE under two additional random seeds ($\{123, 456\}$, matching the 20-epoch primary protocol) confirms that $\hat{\pi}_0$ and BH rejection rates are stable across initializations (coefficient of variation below 6% for $\hat{\pi}_0$ and below 2% for BH rejection rate across all three models), indicating that single-seed FDR comparisons are reliable and the observed cross-model differences are driven by model architecture rather than random initialization. Extending multi-seed evaluation to ComplEx and to FB15k-237 would further strengthen these conclusions. Fifth, the current evaluation does not include a real-world deployment case study. While benchmark results establish the framework’s statistical properties, evaluating its utility within a concrete knowledge-based application—such as clinical decision support, drug repurposing, or financial risk

monitoring—would strengthen the argument that FDR-based reliability assessment translates to practical impact. We consider all five directions important for follow-up work.

4.8. Sensitivity Analysis

To assess the robustness of our FDR framework to its two key hyperparameters—the number of negative samples K and Storey’s tuning parameter λ —and to the choice of FDR control procedure under dependence, we conducted a sensitivity analysis on the WN18RR benchmark.

Sensitivity to K . Table 4 reports $\hat{\pi}_0$ and BH rejection counts $R(0.05)$ for $K \in \{25, 50, 100\}$. For TransE, $\hat{\pi}_0$ is identical at $K = 50$ and $K = 100$ (0.295), while RotatE shows modest variation (0.265 at $K = 50$ vs. 0.259 at $K = 100$, a difference of 0.006). BH rejection counts are sensitive at very small K ($K = 25$ produces $R = 0$ for both models due to insufficient granularity of p -values at $1/26 \approx 0.038$ for $\alpha = 0.05$) but stabilize by $K = 50$. RotatE’s $R(0.05)$ increases from 505 ($K = 50$) to 1,100 ($K = 100$), reflecting the finer p -value resolution at larger K , while TransE plateaus at $K = 50$. The recommended choice $K = 100$ provides a conservative margin above the convergence threshold while remaining computationally accessible.

Table 4: Sensitivity of FDR metrics to the number of negative samples K . $\lambda = 0.5$ fixed.

Model	K=25	K=50	K=100
$\hat{\pi}_0$ (Storey null proportion)			
TransE	0.317	0.295	0.295
RotatE	0.285	0.265	0.259
$R(0.05)$ (BH rejections at $\alpha = 0.05$)			
TransE	0 (0.0%)	262 (9.0%)	262 (9.0%)
RotatE	0 (0.0%)	505 (17.3%)	1,100 (37.6%)

Sensitivity to λ . Table 5 reports $\hat{\pi}_0$ for Storey’s tuning parameter $\lambda \in \{0.3, 0.5, 0.7\}$. The estimated $\hat{\pi}_0$ varies by at most 0.076 across this range, and

the BH rejection count $R(0.05)$ is invariant to λ (since BH does not depend on $\hat{\pi}_0$). The q -value rejection counts vary from 1,843 to 2,050 (TransE) and 2,050 to 2,050 (RotatE) across λ values—a range of approximately 10%—which does not affect any of the paper’s qualitative conclusions.

Table 5: Sensitivity of $\hat{\pi}_0$ to Storey’s tuning parameter λ . $K = 100$ fixed.

Model	$\lambda = 0.3$	$\lambda = 0.5$	$\lambda = 0.7$
TransE	0.330	0.295	0.270
RotatE	0.296	0.259	0.220

Robustness to dependence structure: BY and adaptive BH procedures. The BH procedure guarantees FDR control under the PRDS condition (Section 3.3.1). To assess whether the core findings are robust to violations of this condition, we applied two alternative procedures that relax or eliminate the PRDS requirement.

BY procedure. The Benjamini-Yekutieli (BY) procedure [37] provides FDR control under arbitrary dependence at the cost of reduced power. The BY threshold is $\alpha/(m \cdot H_m)$ where $H_m = \sum_{i=1}^m 1/i$ is the m -th harmonic number; with $m = 2,924$, $H_m \approx 8.29$, making the per-hypothesis threshold approximately 165× more stringent than BH. Consequently, BY produces zero rejections for all four models on both WN18RR ($R_{\text{BY}}(0.05) = 0$) and FB15k-237 at $\alpha = 0.05$ —consistent with the known conservatism of BY when m is large [37].

Two-stage adaptive BH. When $\pi_0 < 1$, the two-stage adaptive BH procedure [38] recovers power by first estimating m_0 and then applying BH at an adjusted threshold $\alpha' = \alpha \cdot m/\hat{m}_0$. This procedure retains finite-sample FDR control at level α under independence, without requiring the PRDS condition to be verified. Table 6 reports adaptive BH rejection counts alongside standard BH for all four models on WN18RR. Adaptive BH substantially increases power (e.g., TransE: 1,447 $\rightarrow$ 1,660 at $\alpha = 0.05$; RotatE: 1,100 $\rightarrow$ 1,916) while preserving the same cross-model ordering (TransE $>$ RotatE $>$ ConvE $>$ ComplEx), confirming that the paper’s comparative conclusions are not artifacts of the

PRDS assumption. ComplEx remains at zero rejections under all procedures, consistent with its $\hat{\pi}_0 = 0.754$.

Table 6: Comparison of BH, BY, and two-stage adaptive BH rejection counts on WN18RR ($K = 100$, $\alpha = 0.05$). Adaptive BH applies the Benjamini-Krieger-Yekutieli (2006) procedure, which estimates $m_0 = m \cdot \hat{\pi}_0$ and adjusts the BH threshold accordingly.

Model	BH	BY	Adaptive BH	$\hat{\pi}_0$
TransE	1,447 (49.5%)	0 (0.0%)	1,660 (56.8%)	0.283
RotatE	1,100 (37.6%)	0 (0.0%)	1,916 (65.5%)	0.259
ComplEx	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.754
ConvE	742 (25.4%)	0 (0.0%)	1,480 (50.6%)	0.321

These results indicate that the paper’s core findings are robust to the choice of FDR control procedure: the existence of a non-trivial null proportion ($\hat{\pi}_0 \gg 0$) and the per-relation heterogeneity are established independently of the specific method, and the cross-model BH ordering (TransE > RotatE > ConvE > ComplEx) is preserved under adaptive BH.

5. Conclusion

We introduced a false discovery rate (FDR) control framework for the evaluation of knowledge graph completion models. By recasting link prediction as a large-scale multiple hypothesis testing problem, the framework provides principled statistical confidence measures for individual predictions—a dimension of evaluation that the standard MRR/Hits@ K paradigm leaves unaddressed.

Experiments on WN18RR and FB15k-237 with TransE, RotatE, ComplEx, and ConvE yielded three findings relevant to the KG completion community. First, the estimated null proportion ranges from $\hat{\pi}_0 = 0.259$ (RotatE, WN18RR) to 0.851 (ComplEx, FB15k-237), indicating that the fraction of test triples indistinguishable from noise varies widely across models and benchmarks: on the larger FB15k-237 benchmark, even the best model (RotatE) has $\hat{\pi}_0 =$

0.582, meaning the majority of test triples carry no detectable signal. Second, per-relation FDR rejection rates exhibit extreme dispersion—from below 10% to above 95% on WN18RR, and from 0.8% to 84.3% across 237 FB15k-237 relations—and this heterogeneity persists across model architectures, consistent with the existence of “knowledge blind spots” that resist embedding-based prediction. Third, the ranking of models by MRR and by FDR power does not fully align: RotatE and TransE each lead on one evaluation dimension, and TransE’s BH rejection rate exceeds ConvE’s by a substantial margin on both benchmarks despite ConvE’s higher MRR on FB15k-237. This finding highlights that ranking quality and statistical reliability are distinct model properties that should both inform deployment decisions in knowledge-based systems.

These findings call into question the assumption that a single ranking metric adequately characterizes model quality. Our results suggest that the KG completion community would benefit from adopting FDR-based diagnostics alongside MRR and Hits@ K as a standard component of model evaluation. At minimum, we recommend reporting $\hat{\pi}_0$ and BH rejection counts at $\alpha = 0.05$, which together quantify the proportion of unpredictable test triples and the absolute number of statistically reliable discoveries—information that ranking metrics alone cannot supply. The practical significance extends beyond benchmarking: for knowledge-based applications that consume KG completion output—drug repurposing, clinical decision support, financial risk monitoring—knowing which predictions are statistically reliable directly informs resource allocation.

The framework is compatible with major embedding paradigms, operates as a post-hoc protocol requiring no model modification, and completes in minutes on consumer hardware. We have released the complete source code, trained model checkpoints, and per-relation FDR statistics to facilitate community adoption.

Several limitations warrant further investigation. The effect of dependence among p -values on FDR control guarantees, while partially addressed through the Benjamini-Yekutieli procedure (see Sensitivity Analysis, §4.8), merits formal theoretical analysis for general KG embedding architectures. A supplementary

multi-seed analysis confirms that $\hat{\pi}_0$ and BH rejection rates are stable across random initializations (COV < 6% and < 2%, respectively, across three model architectures on WN18RR); extending this analysis to FB15k-237 would further quantify cross-benchmark variance. Extending the analysis to larger benchmarks (YAGO3-10, Wikidata5M), additional model families (GNN-based, hyperbolic, LLM-enhanced), and real-world deployment case studies would further establish the generality and practical impact of FDR-based KG evaluation.

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Data Availability Statement

The WN18RR and FB15k-237 datasets used in this study are publicly available benchmark datasets. The complete source code, analysis scripts, trained model checkpoints, and per-relation FDR statistics are openly available at <https://github.com/Jackxiaozhir>. No new data were generated for this study. All experiments were conducted with random seed 42; re-running with the provided code and seed will reproduce the exact results reported in this paper.

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Algorithm 1 FDR-Based KG Completion Evaluation Protocol

Require: Trained KG embedding model with scoring function $\hat{g}$; Test set

$\mathcal{T}_{\text{test}} = \{(h_i, r_i, t_i^*)\}_{i=1}^m$; Entity set $\mathcal{E}$; number of negative samples K ; FDR level α

Ensure: p -values $\{\hat{p}_i\}_{i=1}^m$; BH rejections $R(\alpha)$; Storey $\hat{\pi}_0$, q -values; local FDR

```
1: // Phase 1:  $p$ -value construction
2: for  $i = 1$  to  $m$  do
3:   Sample  $K$  negative tails  $\{t_k^{(i)}\}_{k=1}^K \subset \mathcal{E}$  (filtered)
4:   Compute  $s_i^* = \hat{g}(h_i, r_i, t_i^*)$  ▷ True triple score
5:   for  $k = 1$  to  $K$  do
6:     Compute  $s_{i,k} = \hat{g}(h_i, r_i, t_k^{(i)})$  ▷ Negative triple score
7:   end for
8:    $\hat{p}_i \leftarrow \frac{|\{k: s_{i,k} \geq s_i^*\}| + 1}{K + 1}$  ▷ Eq. 2
9: end for
10: // Phase 2: Benjamini-Hochberg FDR control
11: Sort  $\{\hat{p}_i\}_{i=1}^m$  ascending:  $\hat{p}_{(1)} \leq \dots \leq \hat{p}_{(m)}$ 
12:  $k^* \leftarrow \max\{k : \hat{p}_{(k)} \leq \frac{k}{m} \cdot \alpha\}$ 
13:  $R(\alpha) \leftarrow k^*$  ▷ Number of discoveries at FDR level  $\alpha$ 
14: // Phase 3: Storey's  $q$ -value and  $\hat{\pi}_0$  estimation
15: for  $\lambda \in \Lambda$  do ▷ Grid of tuning parameters, e.g.,  $\Lambda = \{0.1, \dots, 0.9\}$ 
16:    $\hat{\pi}_0(\lambda) \leftarrow \frac{|\{i: \hat{p}_i > \lambda\}|}{m \cdot (1 - \lambda)}$  ▷ Eq. 4
17: end for
18:  $\hat{\pi}_0 \leftarrow \text{smooth}\{\hat{\pi}_0(\lambda)\}_{\lambda \rightarrow 1}$  ▷ Cubic spline extrapolation
19: Compute  $q$ -values from  $\{\hat{p}_i\}$  and  $\hat{\pi}_0$ 
20: // Phase 4: Efron's local FDR
21:  $z_i \leftarrow \Phi^{-1}(1 - \hat{p}_i)$  for all  $i$  ▷ Eq. 6
22: Estimate mixture density  $f(z)$  via kernel density estimation
23:  $\text{fdr}(z_i) \leftarrow \frac{\hat{\pi}_0 \cdot \phi(z_i)}{f(z_i)}$  for all  $i$  ▷ Eq. 8
24: return  $\{\hat{p}_i\}, R(\alpha), \hat{\pi}_0, \{q_i\}, \{\text{fdr}(z_i)\}$ 
```
